

AI Glossary

A comprehensive visual handbook of 30 essential terms, architectures, and methodologies shaping modern Artificial Intelligence & Generative Systems.

How to use this glossary: As Artificial Intelligence transitions from research labs into core software development and enterprise workflows, understanding its foundational terminology is crucial. This guide breaks down key concepts across 5 critical domains: System Architecture, Prompting, Information Retrieval, MLOps, and Training Paradigms.

1. RETRIEVAL, DATA & ARCHITECTURE

Retrieval-Augmented Generation RAG

A technique that enhances Large Language Models by connecting them to external databases or document repositories. Instead of relying solely on pre-trained knowledge, RAG fetches relevant private or up-to-date facts before generating a response.

Context: Used in enterprise chatbots to answer questions based on internal company PDFs without re-training the model.

Vector Database

A specialized database designed to store and query high-dimensional vector embeddings. It allows lightning-fast similarity searches based on semantic meaning rather than exact keyword matches.

Key Tools: Pinecone, Weaviate, Qdrant, Milvus, pgvector.

Vector Embedding

A numerical representation of text, images, or audio as arrays of numbers (vectors) in a multi-dimensional space. Words or sentences with similar meanings are positioned close to each other.

Example: "King - Man + Woman" mathematically points closely to "Queen" in vector space.

Large Language Model LLM

A massive deep learning model trained on vast amounts of text data to understand, generate, and process human language using probabilistic prediction.

Transformer Architecture

The groundbreaking deep learning neural network architecture introduced in 2017 (via the paper "Attention Is All You Need") that powers modern LLMs through parallelized self-attention mechanisms.

Mixture of Experts MoE

A machine learning technique where multiple specialized sub-networks ("experts") handle different parts of the input data, with a gating network deciding which expert to activate per token, lowering computational costs.

2. PROMPT ENGINEERING & INFERENCE CONTROLS

Prompt

The textual or multimodal input provided to an AI model to guide its output. A prompt sets the task, tone, role, constraints, and context for generation.

Prompt Engineering

The practice and discipline of designing, structuring, and optimizing prompts to programmatically elicit accurate, consistent, and safe outputs from generative AI models.

Context Window

The maximum amount of information (measured in tokens) that an AI model can process and hold in memory during a single conversation or inference request.

Impact: A 128k token context window allows an AI to read an entire book in a single prompt.

Token

The basic unit of text processed by an LLM. A token can be a word, part of a word, or punctuation. On average, 1,000 tokens equal roughly 750 English words.

Temperature

A hyperparameter that controls the randomness and creativity of an AI's output. Lower temperatures (e.g., 0.1) produce deterministic and focused responses, while higher temperatures (e.g., 0.8) yield creative and diverse outputs.

Zero-Shot / Few-Shot Learning

Zero-Shot: Asking the AI to perform a task without giving explicit examples.

Few-Shot: Providing a few sample input-output pairs inside the prompt to guide output structure and style.

AI Guardrails

Safety protocols, rules, and moderation layers wrapped around an AI system to ensure outputs remain compliant, accurate, secure, and non-toxic.

3. SOFTWARE DEVELOPMENT & OPERATIONS

AI-Driven Development

A software engineering paradigm where AI assistants (e.g., GitHub Copilot, Cursor) are deeply embedded into the coding lifecycle, assisting in code generation, refactoring, testing, and debugging.

Machine Learning Operations MLOps

A set of practices that combines Machine Learning, DevOps, and Data Engineering to deploy, monitor, automate, and maintain ML models in production reliably and efficiently.

Agentic AI / AI Agent

An autonomous AI system that can reason, break down complex goals into smaller steps, execute commands using external tools/APIs, and self-correct to complete tasks with minimal human intervention.

Inference

The operational phase where a trained AI model processes new, unseen user input to generate predictions or outputs in real time.

Fine-Tuning

The process of taking a pre-trained base model and further training it on a smaller, domain-specific dataset to adapt its style, behavior, or domain knowledge.

Quantization

A model optimization technique that compresses deep learning models by lowering the precision of their weights (e.g., from 16-bit floats to 4-bit integers), allowing them to run on hardware with less memory.

4. MODEL RELIABILITY, SAFETY & EVALUATION

Hallucination

A phenomenon where a generative AI model confidently outputs false, fabricated, or factually incorrect information presented as absolute truth.

AI Alignment

The research area focused on ensuring that AI systems act in accordance with human values, safety criteria, ethical standards, and user intentions.

Perplexity

A mathematical metric used to evaluate language models. It measures how well a probability model predicts a sample text. Lower perplexity indicates higher model confidence and accuracy.

Overfitting

A training flaw where a machine learning model memorizes its training data too closely, performing exceptionally on training data but failing on new, unseen input.

5. TRAINING PARADIGMS & GENERATIVE MODELS

Supervised Learning

A foundational machine learning approach where a model is trained on labeled data (input paired with the correct output) to learn classification or regression tasks.

Unsupervised Learning

Training an AI model on unlabeled data, allowing the algorithm to discover hidden patterns, clusters, or structures on its own.

Reinforcement Learning from Human Feedback RLHF

A fine-tuning method where human evaluations and rankings are used to train a reward model, which then optimizes the AI model to produce safer, more helpful responses.

Multimodal AI

An AI system capable of processing and understanding multiple types of input modalities simultaneously, such as text, images, video, and audio.

Diffusion Model

A class of generative models primarily used for image generation (e.g., Midjourney, Stable Diffusion) that learn to generate data by gradually removing noise from a completely random image signal.

Latent Space

A lower-dimensional vector space in which a neural network maps complex input data (like images or sentences) to uncover intrinsic features and abstract relationships.